1. **Scientific Question**: How will concentration of transcription factors (TF) change the rate of protein production in prokaryotes?

2. **Product**: The product is a simulator of gene expression on three levels: mRNA transcription regulation, mRNA production, and protein production.

The function of activators and repressors of transcription level will be based on this formula:

$$r(x) = r_{basal} + (r_{max} - r_{basal}) \prod_{i=1}^{m} h_i(x_i)$$

$$h_i(x_i) = \begin{cases} \dfrac{{K_i}^{n_i}}{{K_i}^{n_i} + {x_i}^{n_i}} \; when\; factor\; is\; repressor \\ \dfrac{{x_i}^{n_i}}{{K_i}^{n_i} + {x_i}^{n_i}} \; when\; factor\; is\; activator \end{cases}$$

For convenience of calculations, we will use $r_{max} = 1$ and $r_{basal} = 0.001$ in this formula. The user will have to type in the K value and n values by themselves. User can superimpose the functions in one x-y plane to compare the effect of different transcription factors.

The calculation of mRNA concentration uses this formula:

$$\frac{dm}{dt} = r(x) - \gamma_m m$$

The calculation of protein production uses this formula:

$$\frac{dp}{dt} = \beta m - \gamma_p p$$

Moreover, Gillespie's algorithm will also be added into the simulation of this model to add some noise into the overall reaction.

3. **Test**: I will be using special cases of the formula to check the correctness of the value in the three levels.

Default test values in first level: r_{max} = 1, r_{basal} = 0.001, K = 10, n = 2. For the repressor case, the formula should be returning a 1 when x = 0, about 0.5005 when x = K, and 0.0011 when x = 1000. For the activator case, the return values for x = 0 is 0.001, and 1 when x = 1000. The x = K case has the same return values with the repressor case. Both functions reject negative x and negative rates, nonpositive K and n values, and $r_{basal} >= r_{max}$.

For the second and third level, with r = 1, γ_m = 0.5, β = 1, and γ_p= 0.5, m = 2, p = 4. If β = 0, protein is 0, while mRNA is 2.

The three levels will then be tested using respectively these three papers: Golding et al. 2005, Cell; Garcia & Phillips 2011, PNAS; Golding et al. 2005, Cell; Thattai & van Oudenaarden 2001, PNAS; Cai et al. 2006, Nature; Elowitz et al. 2002, Science. My model needs to replicate the results of these three questions.